

Course Proposal

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The Hardware Underground: Firmware & Hardware Security Engineering

Modern critical infrastructure, automotive systems, industrial control systems, IoT devices, routers, medical equipment, and consumer electronics all rely on embedded systems and firmware.

This course provides students with the fundamental and advanced knowledge required to analyze, assess, secure, and understand embedded devices at both software and hardware levels.

Students will acquire practical skills in:

- Reverse Engineering
- Vulnerability Analysis
- Exploit Development
- Firmware Security
- Hardware Hacking
- Secure Embedded Design
- Hardware Root of Trust
- Wireless Protocol Security

The course combines theoretical instruction with hands-on laboratory exercises using real hardware and firmware images.

Modules*

All course materials, including presentations, laboratory guides, and examples, will be published through a dedicated Git server repository. The course will primarily use open-source tools and Linux-based virtual machines, allowing students to complete the initial modules using only standard computing equipment, with specialized hardware introduced in later hands-on laboratories.

To support collaboration and practical learning, a dedicated discussion platform and a customized CTF infrastructure are planned. High-performing students will be recognized through advanced challenge tracks, with custom-designed hardware badges awarded as a distinction for exceptional achievement.

**NOTE: all written topics are subject to change. The overview is there to only identify the initial project goal*

Module 0x00 - Computer Architecture Foundations

Topics:

- Binary and hexadecimal systems
- CPU architecture
- Memory organization
- Registers
- Stack and heap
- Calling conventions
- Virtual memory
- Interrupts
- Cache hierarchy

Laboratory:

- Memory inspection
- Process analysis
- Stack visualization

Module 0x01 - Secure Systems Programming

Topics:

- Pointers
- Memory management
- Structures
- Bit operations
- Memory corruption fundamentals
- Unsafe programming patterns

Laboratory:

- Pointer analysis
- Vulnerable application studies

Module 0x02 - Assembly Language Fundamentals

Topics:

- x86-64 Assembly
- ARM Assembly
- Function calls
- System calls
- Control flow

Laboratory:

- Debugging exercises
- Binary tracing

Module 0x03 - Reverse Engineering

Topics:

- Static Analysis
- Dynamic Analysis
- Control Flow Analysis
- Protocol Reverse Engineering
- Malware Analysis Fundamentals

Laboratory:

- Reverse engineering of software and firmware samples

Module 0x04 - Vulnerability Research

Topics:

- Buffer Overflows
- Integer Overflows
- Format String Bugs
- Use-After-Free
- Race Conditions

Laboratory:

- Vulnerability discovery exercises

Module 0x05 - Exploit Development

Topics:

- Stack Exploitation
- Return-Oriented Programming
- Mitigation Bypasses
- Embedded Exploitation

Laboratory:

- Proof-of-concept exploit construction

Module 0x06 - Firmware Security

Topics:

- Firmware Architecture
- Bootloaders
- Linux Kernel
- Device Tree
- Filesystems

Laboratory:

- Firmware extraction and analysis

Module 0x07 - Electronics Fundamentals

Topics:

- Voltage/Current/Resistance
- Basic Components
- Digital Electronics
- Soldering and Rework

Laboratory:

- Embedded Linux security review

Module 0x08 - Hardware Interfaces

Topics:

- UART
- SPI
- I2C
- JTAG/SWD

Laboratory:

- Hardware identification
- Flash extraction
- Debug interface analysis

Module 0x09 - Secure Boot & Hardware Trust

Topics:

- Secure Boot
- TPM
- Hardware Root of Trust
- TrustZone
- Secure Elements

Laboratory:

- Secure boot assessment

Module 0x0A - Hardware Attacks

Topics:

- Fault Injection
- Voltage Glitching
- Clock Glitching
- Side-Channel Analysis
- Power Analysis

Laboratory:

- Controlled fault injection demonstrations

Module 0x0B - Wireless Security

Topics:

- Bluetooth Low Energy
- Zigbee
- LoRa
- RFID
- NFC

Laboratory:

- Protocol analysis
- Traffic inspection

Laboratory Equipment Requirements

Required Hardware:

- Raspberry Pi 5
- STM32 Development Boards
- BeagleBone Black
- Logic Analyzer
- Oscilloscope
- UART Adapters
- JTAG/SWD Debuggers
- RTL-SDR
- ESP32 Development Boards
- Assorted IoT Devices

Software:

- Linux
- Radare2
- Binwalk/EMBA
- QEMU
- OpenOCD/GDB
- Wireshark
- SDR++/URH